

Prediction of catheter's tip position based on X-ray images of renal-disease patients subjected to dialysis

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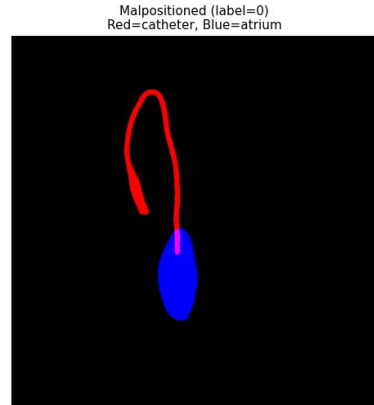
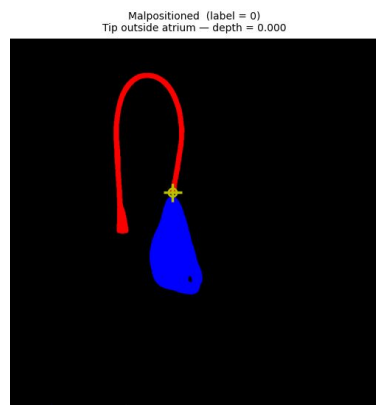
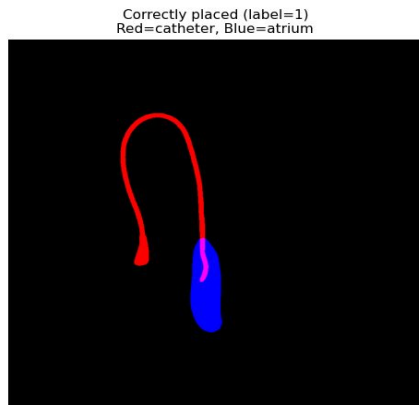
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Introduction: The Clinical Problem

Central Venous Catheters (CVC)

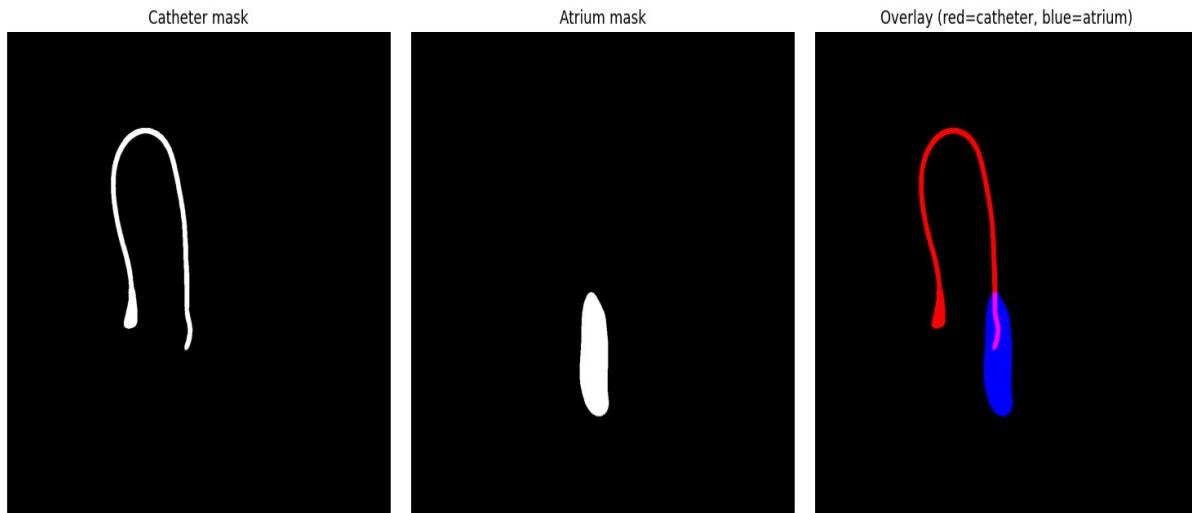
- Life-supporting devices for dialysis patients.
- Tip must sit within the correct depth zone of the right atrium - being inside the atrium alone does not guarantee correct placement.
- Malposition causes:
 - Cardiac arrhythmia
 - Inadequate dialysis flow
- Radiograph review is standard but slow and variable.
- Automated pipeline could reduce delay between insertion and verified safe use



Prior Work

Forgwa (2025) - Baseline

- UNet++ segmentation pipeline with hybrid Dice + Focal loss to produce binary masks of the catheter and right atrium from the chest X-rays.
- Fed these masks to an EfficientNet-B7 classifier.
- Designed a three-channel input representation- catheter mask, atrium mask, and zeros placeholder.



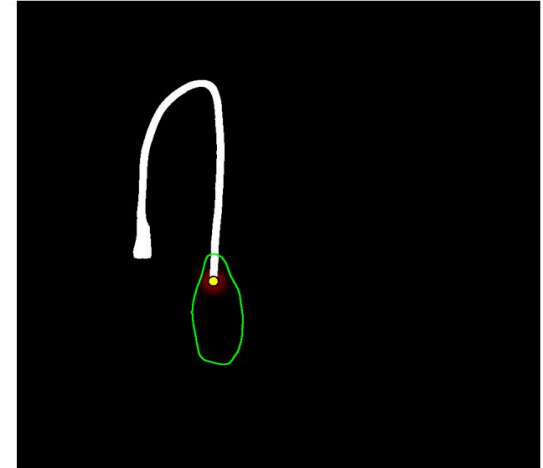
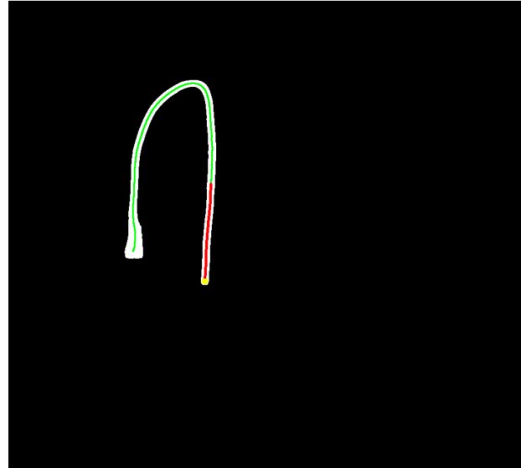
Research Question

RQ1 Can skeleton-based tip detection reliably locate the catheter free endpoint?

RQ2 Do tip-point features provide discriminative power for classical ML?

RQ3 Does a tip-dot EfficientNet-B7 outperform classical ML?

RQ4 Does a two-step classification strategy improve overall classification?



Dataset

Dataset Overview

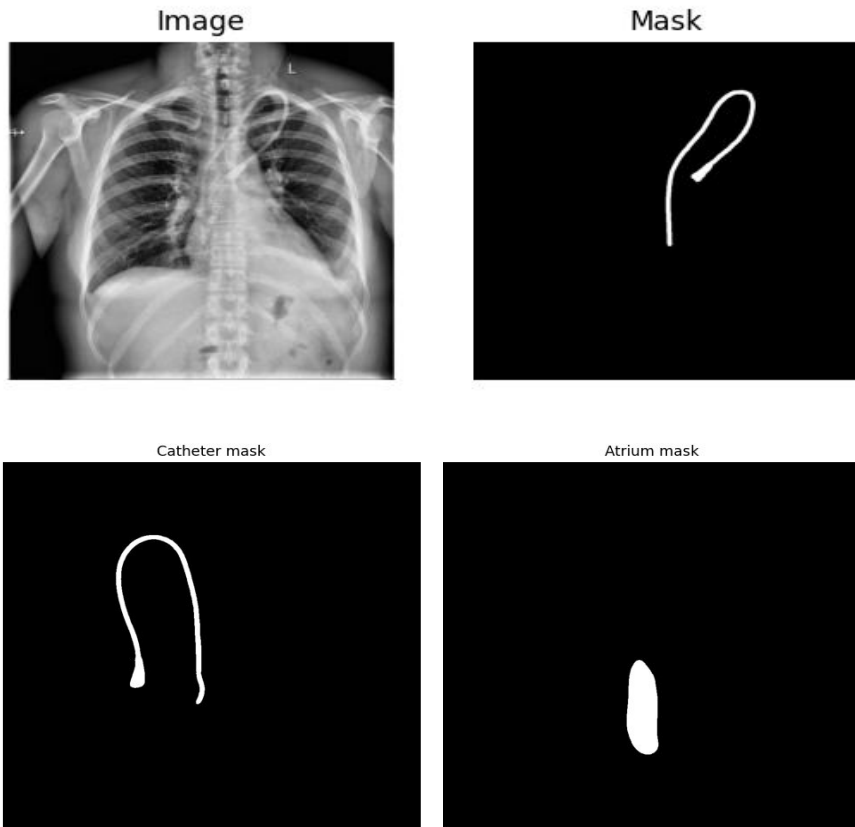
- $n = 220$ posterior-anterior chest X-rays
- Source: Bialystok Medical University, Poland
- Labels assigned by radiologists at the hospital

Class Distribution

- Malpositioned (label = 0): 119 (54.1%)
- Correctly placed (label = 1): 101 (45.9%)

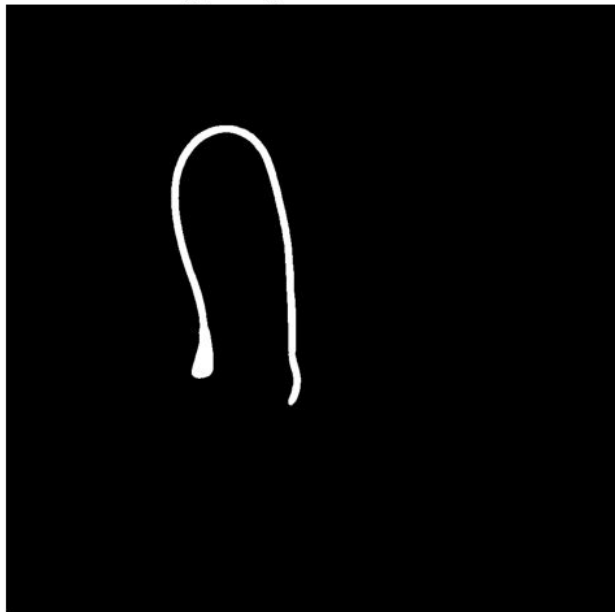
Input representation

- UNet++ predicted binary segmentation masks
- 960 x 960 grayscale PNG per structure
- Use Stratified 5-fold cross validation

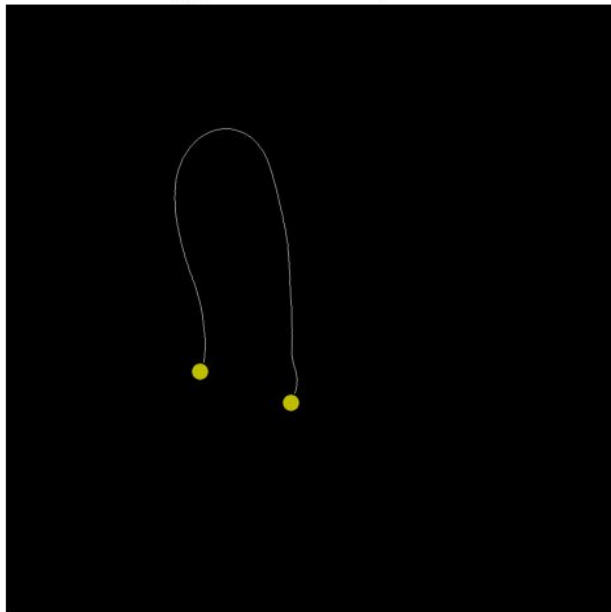


Tip Detection Algorithm

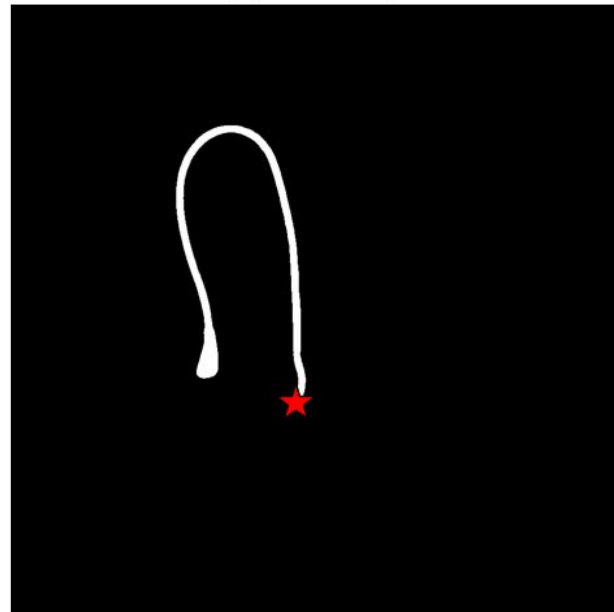
(a) Binary catheter mask



(b) Skeleton + endpoints



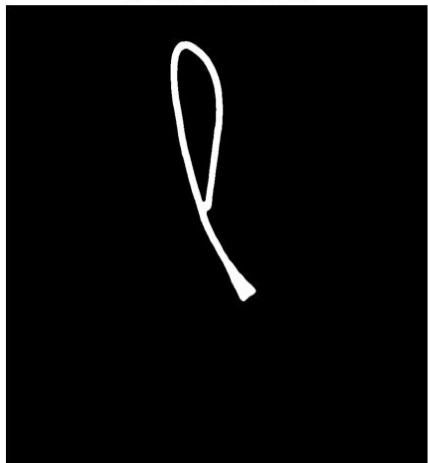
(c) Detected tip



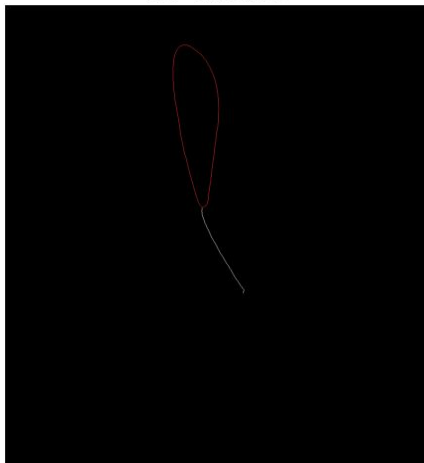
Loop Recovery

Loop detection and recovery — IMG-0179-00001.png

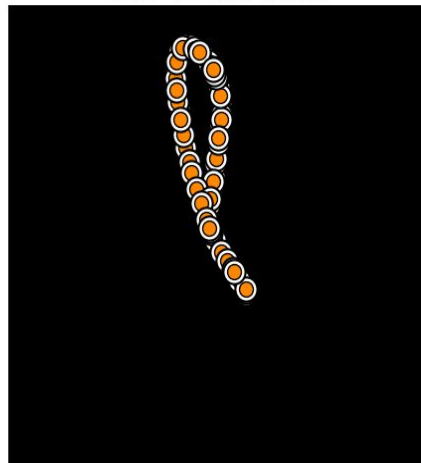
(a) Binary catheter mask



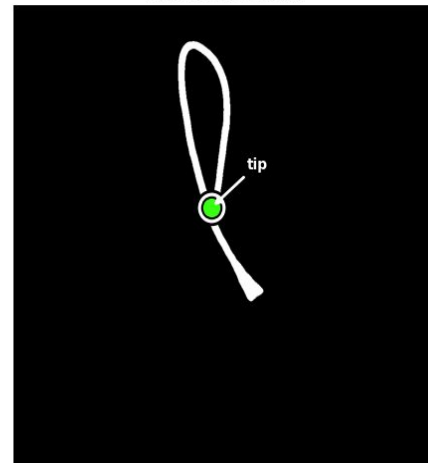
(b) Looped skeleton
Red = cycle pixels



(c) After junction cut
Circles = recovered endpoints



(d) Detected tip
(after loop recovery)



Tip Detection Status

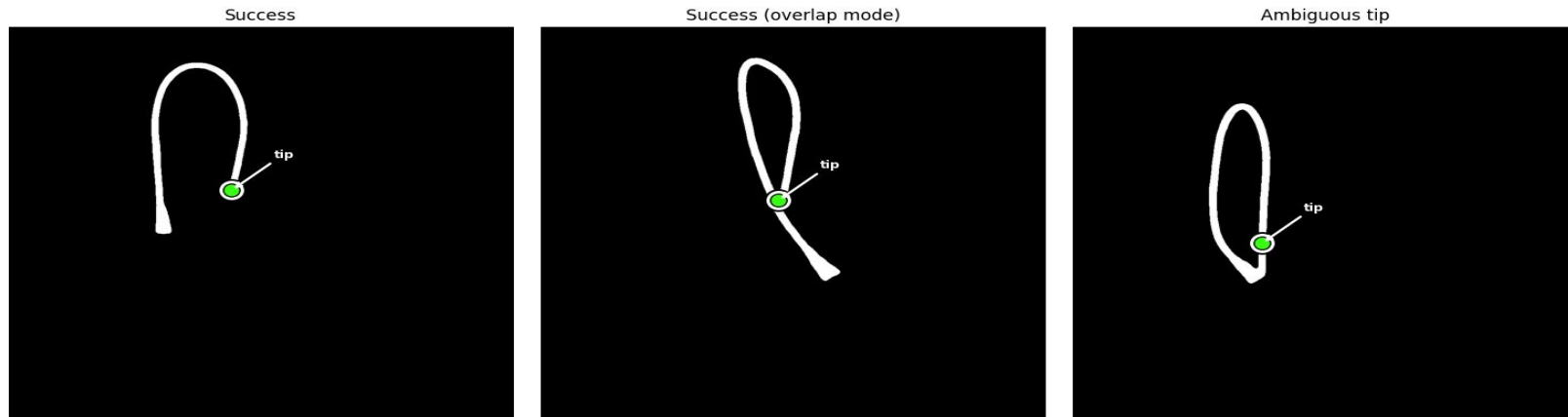


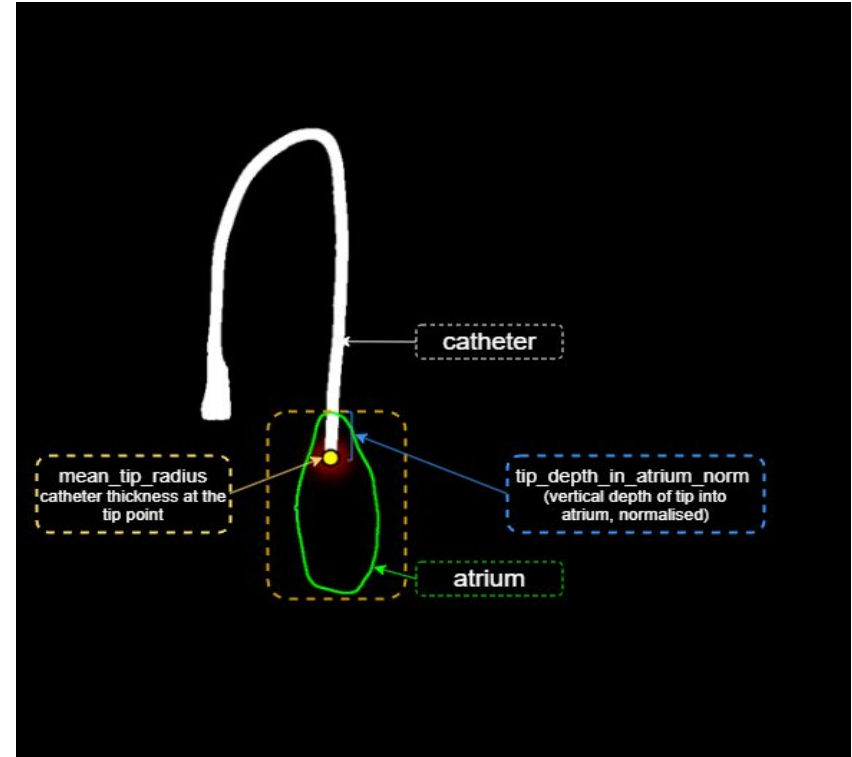
Table 2: Tip detection status distribution ($n = 220$).

Status	Description	n	%
success	Clean catheter; tip found via thickness rule	197	89.5
success_overlap_mode	Loop recovered; tip found after cutting	3	1.4
ambiguous_tip	Recovery failed; tip unreliable	20	9.1

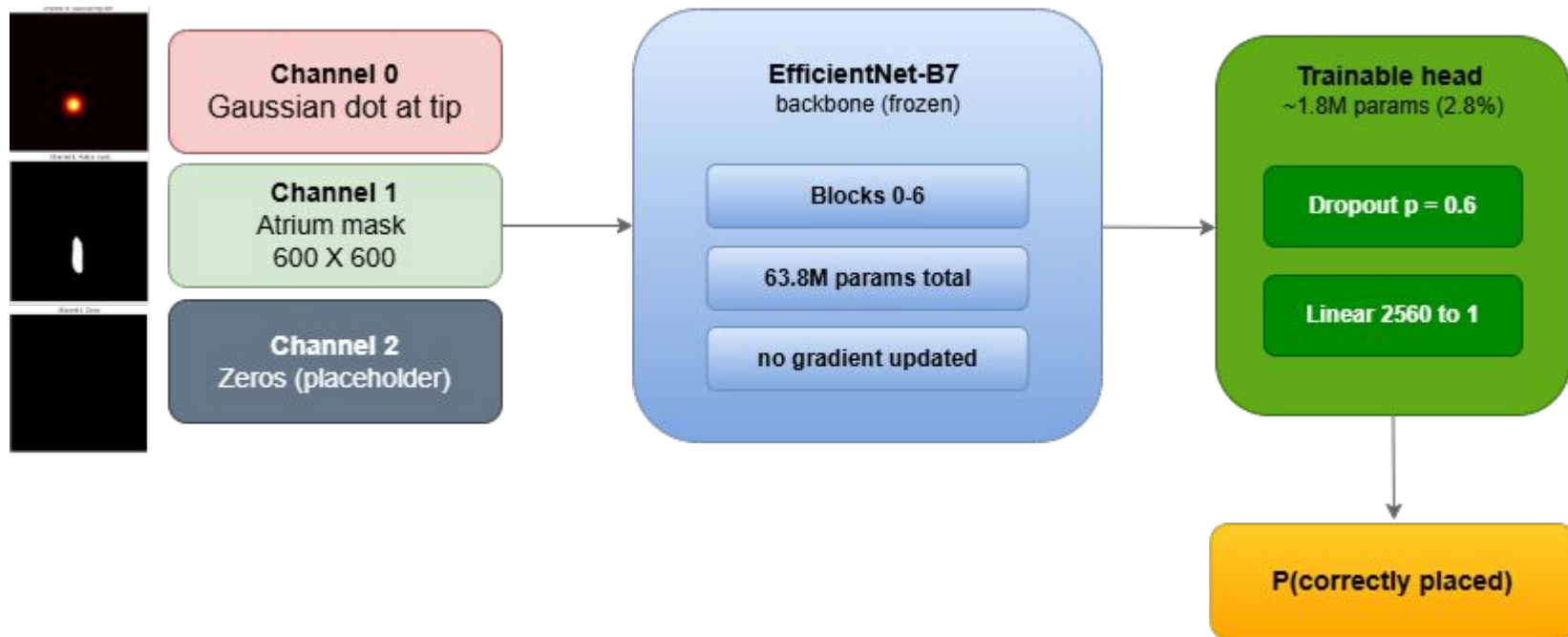
Feature Engineering

F1 Arc-length depth of the tip inside the atrium (0 = top, 1 = bottom)

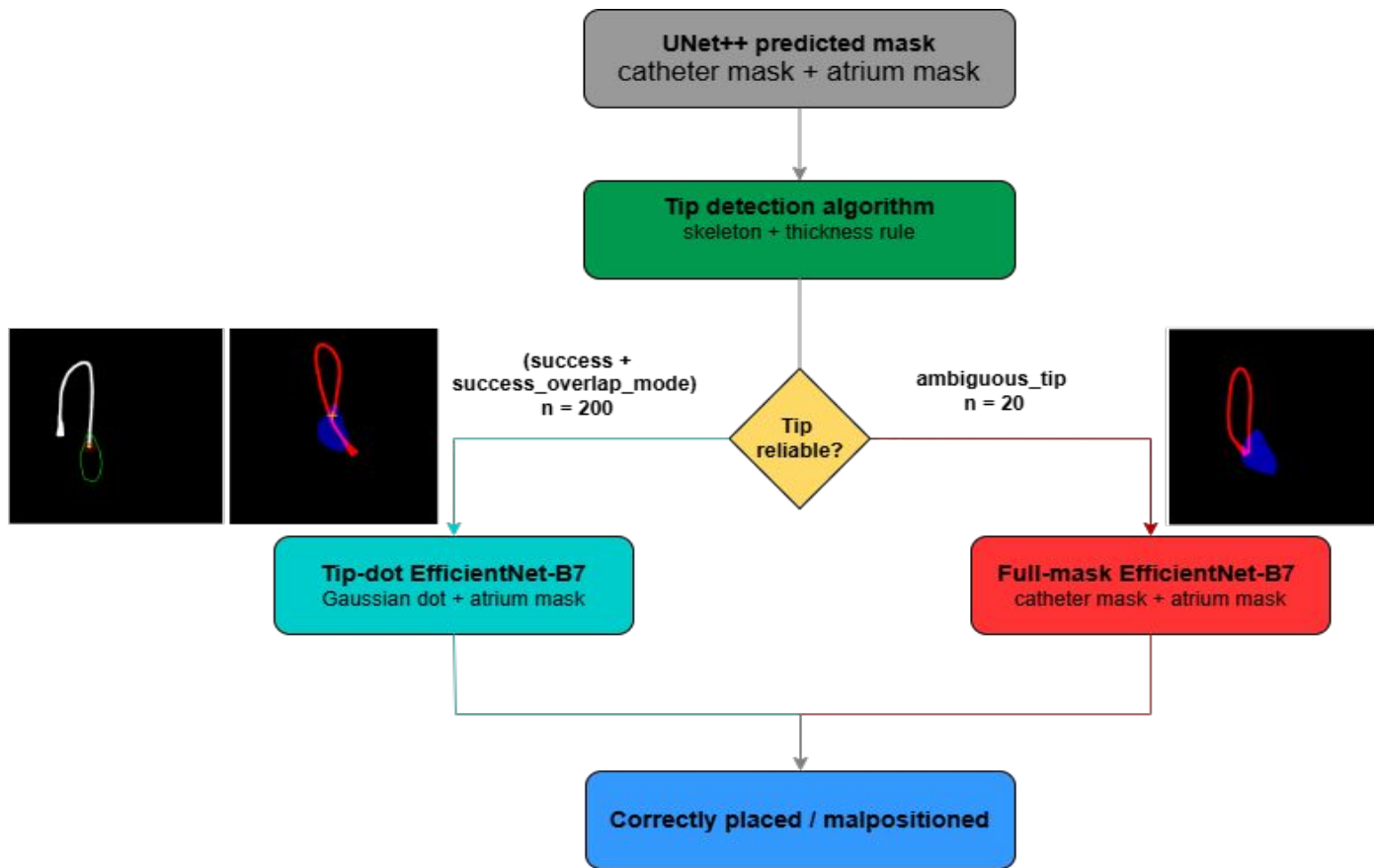
F2 Thickness of the catheter endpoint.



Tip-Dot EfficientNet-B7



Two-Step Classifier



Full Comparison and Improvement Summary

	Method	AUC	Accuracy	Sensitivity	Specificity	F1
ML	Logistic Regression	0.833	0.814	0.782	0.840	0.794
	Random Forest	0.900	0.795	0.792	0.798	0.780
	XGBoost	0.881	0.823	0.772	0.866	0.800
DL	EfficientNet-B7 tip-dot (proposed)	0.892	0.855	0.832	0.874	0.840
	EfficientNet-B7 full mask (Forgwa, 2025)	0.930	0.859	0.890	0.832	0.853
	Two-step classifier (proposed)	0.965	0.914	0.931	0.899	0.908

Limitations

Limitations

- **Small single-centre dataset**

220 images from one hospital

- **Segmentation Dependency**

9.1% ambiguous tips when masks poor.

- **Frontal images only**

Lateral view depth info not available

- **Inter-rater variability**

No inter-annotator agreement measure

- **Limited architectures**

Only EfficientNet-B7 explored

Future Work

Future Work

- External multi-centre validation
Multiple hospital and scanner types
- End-to-end training
Jointly optimise tip localisation + classification
- Uncertainty quantification
Confidence intervals for cases needing human review
- Real-time deployment
DICOM-integrated bedside tool

Ethical Thinking, Societal Relevance & Stakeholder Awareness

Ethical Aspects

- Data pseudonymised - GDPR complaint
- Handled only on secure VSC and Colab infrastructure
- Ethical consent granted by Bialystok Medical University
- All results from genuine cross-validated evaluations
- Single-centre training data may introduce algorithmic bias - requires investigation before deployment.

Societal Relevance

- CVC complications affect ~15% of placement globally
- Automated tool could:
 - Reduce delay between insertion and verified safe use
 - Provide safety net when radiologist unavailable
 - Support less experienced clinicians
- Missed malposition directly harms the patient: specificity is the primary safety metric

Stakeholder Awareness

- Patients: reduced complications and mortality
- Clinicians: decision-support tool not a replacement
- Hospitals: cost savings
- Regulators: CE marking under EU MDR required before deployment
- Researchers: reproducible work

Conclusion

We set out to improve on Forgwa's baseline

- Building on the UNet++ segmentation pipeline.

What we build

- A skeleton-based tip detection algorithm with loop recovery
- Two hand-crafted tip-point features capturing the dominant clinical signal
- A tip-dot EfficientNet-B7 model encoding precise tip location
- A two-step classifier strategy combining both models by reliability

It worked & it improved over the baseline

- AUC 0.965 vs 0.930 - more discriminative overall
- Specificity 0.899 vs 0.832 - fewer correctly placed catheters falsely flagged

Thank You